COMPOUND & COMPILE

The Data Science Learning Roadmap

A realistic 16-week path from zero to your first real project.

1. Foundations (Weeks 1-3)

- Python basics — variables, loops, functions, data structures
- Git & GitHub — version control, basic workflow
- SQL — SELECT, JOIN, GROUP BY, basic query writing

2. Core Data Skills (Weeks 4-7)

- Statistics — distributions, hypothesis testing, p-values
- Pandas & NumPy — data manipulation, cleaning, transformation
- Data visualization — Matplotlib, Seaborn, when to use which chart
- Exploratory Data Analysis — reading a messy dataset properly

3. Machine Learning (Weeks 8-12)

- Regression & classification — the core supervised models
- Model evaluation — accuracy traps, precision/recall, cross-validation
- Feature engineering — turning raw data into usable signals
- scikit-learn — the library you'll use for almost everything here

4. Build & Apply (Weeks 13-16)

- Pick ONE real, messy dataset — not a cleaned Kaggle toy set
- Build a complete project end-to-end: clean -> analyze -> model -> explain
- Write up what you found and why it matters, in plain language
- Put it on GitHub with a clear README — this is your portfolio piece

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Four things that actually matter

- One finished messy-data project beats five tutorial-following exercises.
- Statistics has the highest return on study time of anything on this list.
- SQL shows up in interviews far more than people expect going in.
- Depth on one project beats breadth across ten shallow ones, every time.